

Joongheon Kim

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Professor, Korea University – School of Electrical Engineering, Seoul, Korea

Visiting Professor, Seoul National University Hospital, Seoul, Korea

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Educational Backgrounds

- **University of Southern California (USC) – Viterbi School of Engineering** *Los Angeles, California, USA*
 - Ph.D. (08/2009–08/2014) in **Computer Science**, Mark and Mary Stevens School of Computing and Artificial Intelligence
Advisor: Prof. Andreas F. Molisch (Communications, Information, Learning & Quantum (CLIQ) Group (Electrical Engineering))
 - M.S. (05/2014) in **Computer Science** with specialization in **High Performance Computing and Simulations**
 - M.S. (05/2012) in **Electrical Engineering**
 - **Korea University – College of Informatics** *Seoul, Republic of Korea*
 - M.S. (03/2004–02/2006) in **Computer Science and Engineering**
 - B.S. (03/1999–02/2004) in **Computer Science and Engineering**
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Professional Affiliations

Full-Time

- **Korea University**, *Seoul, Republic of Korea*
 - Professor (03/2026–Present)**, School of Electrical Engineering (Graduate: Department of Electrical and Computer Engineering)
 - Associate Professor (03/2021–02/2026)**, School of Electrical Engineering
 - Assistant Professor (09/2019–02/2021)**, School of Electrical Engineering
 - Adjunct Professor (03/2023–Present)**, Department of Communications Engineering (Samsung Electronics)
 - Adjunct Professor (03/2021–Present)**, Department of Semiconductor Engineering (SK Hynix)
 - Adjunct Professor (11/2022–Present)**, Department of Future Science and Technology Business (Graduate School)
 - Affiliated Professor (03/2025–Present)**, Department of Defense Convergence Technology (LIG Nex1, Graduate School)
- **Chung-Ang University**, *Seoul, Republic of Korea*
 - Assistant Professor (03/2016–08/2019)**, School of Computer Science and Engineering
- **Intel Corporation**, *Santa Clara (Silicon Valley), California, USA*
 - Systems Engineer (09/2013–02/2016)**, Platform Engineering Group
- **University of Southern California – Viterbi School of Engineering**, *Los Angeles, California, USA*
 - Ph.D. Student in Computer Science (08/2009–08/2014)**, Thomas Lord Department of Computer Science
 - Annenberg Graduate Fellow (2009)**, 4-Year Full Scholarship for the Ph.D. Program in Computer Science
 - Research Assistant (01/2011–08/2014)**, Communications, Information, Learning & Quantum (CLIQ) Group
 - Teaching Assistant (01/2012–05/2013)**, Computer Science and Electrical Engineering Major Courses
- **InterDigital**, *San Diego, California, USA*
 - Intern (05/2012–08/2012)**, Wireless Systems Evolution Department
- **LG Electronics – CTO Office**, *Seoul, Republic of Korea*
 - Research Engineer (01/2006–08/2009)**, Multimedia Research Lab, Seocho R&D Campus

Joint Appointments for Quantum Research Collaboration

- **Seoul National University Hospital**, *Seoul, Republic of Korea*
 - Visiting Professor (10/2025–Present)**, Healthcare AI Research Institute

Services for National R&D

- **Presidential Council on National Artificial Intelligence Strategy**, **Advisory Committee (12/2025–Present)**
 - **National Research Foundation of Korea**, **Review Board (11/2024–10/2026)**
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University R&D/Administrative Positions

R&D Leadership Positions

- **Korea University**, *Seoul, Republic of Korea*
 - Director (07/2024–12/2031)**, Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center
 - Principal Investigator (07/2024–12/2031)**, Software Star-Lab (Quantum AI Empowered Second-Life Platform Technology)

Administrative Positions

- **Korea University**, *Seoul, Republic of Korea*
 - Vice Department Chair (01/2025–08/2025)**, Academic Affairs, School of Electrical Engineering
 - Deputy Vice President (02/2022–08/2024)**, Office of Academic Affairs
 - Dean (06/2021–08/2023)**, Center for Teaching and Learning (CTL)

Committee Members

- **Korea University**, Seoul, Republic of Korea
Member (12/2023–Present), Council of the Interdisciplinary Program in Quantum Information Science
Member (03/2021–02/2027), Korea University 120th Anniversary Planning Committee
Member (06/2021–03/2022, 09/2022–08/2023), Academic Affair Planning Committee
Member (03/2021–08/2023), Distance Learning Management Committee
Steering Committee Member (07/2020–06/2022), Institute of Data Science (IDS)

Awards and Honors

Research and Academic Excellence (International)

- **Top Cited Article in ETRI Journal (Wiley), among work published in 2024** 2025
- **Finalist, AAAI Student Abstract and Poster Session – Oral Presentation** 2026 (Top 35), 2023 (Top 25)
- **IEEE ICTC Best Paper Award – IEEE Communications Society** 2022
- **Spotlight, Oral Presentation – ICML Workshop on Dynamic Neural Networks (2022)** 2022
- **IEEE MMTC Best Journal Paper Award – IEEE Communications Society** 2021
- **IEEE ICOIN Best Paper Award – IEEE Computer Society** 2021
- **IEEE MMTC Outstanding Young Researcher Award – IEEE Communications Society** 2020
- **IEEE Systems Journal Best Paper Award – IEEE Systems Council** 2020
- **Next Generation and Standards (NGS) Division Recognition Award – Intel Corporation** Q1/2015
- **Annenberg Graduate Fellowship Award – University of Southern California** 2009–2013
Awarded with Ph.D. Admission: 4 Year Full Scholarship (Tuition Waiver and \$120,000 Stipend (\$30,000/year for 4 years))
- **IEEE Seoul Section Student Paper Contest** Gold (2019), Silver (2025), Bronze (2025, 2024, 2024, 2023, 2020)
- **IEEE VTS Seoul Chapter Award – IEEE Vehicular Technology Society** 2023, 2022, 2022, 2021, 2021, 2019

Research and Academic Excellence (Korea Regional, since 2016)

- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS** 2025
- **Best Paper Award, The Journal of KICS – KICS** 2024
- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS** 2023
- **Haedong Young Scholar Award – KICS and Haedong Foundation** 2018
For recognizing a researcher under the age of 40 who has made outstanding contributions to IT R&D

Korea University

- **Granite Tower Best Research Award (Top 3%), Totally, 2 times**
– 2025 (Top 3%, top 53 professors at Korea University in 2025) 05/2026
– 2024 (Top 3%, top 54 professors at Korea University in 2024) 05/2025
- **Best Research Achievement Award – Korea University, School of Electrical Engineering** 2024
- **Insung Research Grant Award – Korea University** 01/2023
For recognizing professors in top 5% research excellence during the first 3 years at Korea University
- **Granite Tower Best Teaching Award (Top 5%), Totally, 7 times**
– Probability and Random Process KECE209-02 (2025)
– Software Programming Basics GECT002-01 (2024), GECT002-06 (2025)
– Computer Language and Laboratory EGRN151-07 (2019), EGRN151-06 (2021)
– Introduction to Computers SEMI103 (2021)
– Future Mobility Technology GEQR075 (2022)
- **Best Teaching Award (Top 20%), Totally, 7 times**
– Probability and Random Process KECE209-02 (2021), KECE209-02 (2022)
– Software Programming Basics GECT002-02 (2024), GECT002-08 (2024), GECT002-09 (2025)
– Computer Language and Laboratory EGRN151-09 (2020)
– Object-Oriented Programming SEMI104 (2021)

Academic and University Services (International)

- **Certificate of Appreciation – IEEE Future Networks, AI/ML Working Group** 08/2025
- **Certificate of Appreciation – IEEE/IFIP WiOpt (2024)** 10/2024
- **Best Editor Award – ICT Express (Elsevier)** 07/2023
- **2022 Best Chapter Award, IEEE Vehicular Technology Society Chapter, Awarded as a Treasure** 2022
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021
- **Certificate of Appreciation – Department of Computer Science, University of Southern California** 2010

Academic and University Services (Korea Regional)

- **Outstanding Contribution Award**
– The Korean Institute of Communication Sciences and Information Sciences (KICS) 2025, 2024, 2021, 2019
– Korean Institute of Information Scientists and Engineers (KIISE) Information Network Society 2023, 2022
– Open Standards and ICT Association (OSIA) 2025, 2021
– The Institute of Electronics and Information Engineers (IEIE) 2025

- **Commemorative Plaque, KICS Excellent Lecture** (Title: Quantum Machine Learning and Software Technologies) 2025
- **Best Research Group Award, KICS Research Group in Quantum Communications and Quantum Computing** (Group Leader) 2025
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021

Student Awards (International)

- **IEEE TCSC Award for Excellence in Scalable Computing** (IEEE Computer Society), Awarded to *Soohyun Park* 2025
- **Young Scientist Award** (ICSANE 2025), Awarded to *Minji Lee* 2025
- **IEEE Vehicular Technology Society Student Scholarship Award (2025)**, Awarded to *Gyu Seon Kim* 2025
- **IEEE ICDCS Best Runner-Up Poster Paper Award (2025)**, Awarded to *Seok Bin Son, Soohyun Park* 2025

R&D Projects

University-Wide/Center/Flagship Projects

- **(Center Director, PI) Net-Zero CAFE Research Center** (07/2024–12/2031) ITRC (Korea Univ)
- **(SW Star-Lab) Quantum AI Empowered Second-Life Platform Technology** (07/2024–12/2031) Star-Lab (Korea Univ)
- **Intelligent 6G Wireless Access System Research Center** (04/2021–12/2025) 6G AI Research Center (Korea Univ)
- **K-Starlink: Dynamic Reconfigurable and Intelligent Space-Terrestrial Networks** (06/2021–05/2024) Basic Research Lab (Ajou)
- **Nano UAV Intelligence Systems Research Lab** (10/2020–08/2023) ADD Military Special Research Center (Kwangwoon)
- **5G/Unmanned Vehicle Research Center (5G/UV-RC)** (06/2020–12/2022) ITRC (Hanyang)
- **Human Resource Development for the Biomedical Unstructured Big Data Analysis** (08/2018–12/2021) ITRC (SNU-Hospital)
- **Novel Data Science Driven Framework for Efficient Network Design** (06/2017–05/2020) Basic Research Lab (Chung-Ang)
- **Intelligent Internet of Energy (IoE) Data Research Center** (02/2020–05/2020) ITRC (Kookmin)

Consortium Projects

- **6GARROW: 6G AI-Native Integrated RAN-Core Networks** (09/2024–08/2027) IITP
- **AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development** (04/2022–12/2026) IITP
- **Development of Integrated Development Framework that supports Automatic Neural Network Generation and Deployment optimized for Runtime Environment** (04/2021–12/2025) IITP
- **Integrated Perception Technology Developments for Public Safety Platforms** (06/2019–05/2023) NRF
- **Development of Privacy-Reinforcing Distributed Transfer-Iterative Learning Algorithm** (07/2019–12/2021) MHW
- **Virtual Presence in Moving Objects through 5G (PriMO-5G)** (06/2018–06/2021) IITP
- **Distributed Secure Platform for Scalable Clinical OMOP CDM Models** (04/2019–12/2020) MHW

Individual Projects

- **Quantum-Empowered Spatio-Temporal Multi-Scale Digital Twin System** (03/2025–02/2028) NRF
- **Research on Trainability and Learnability in Quantum Machine Learning Models** (03/2026–10/2027) ETRI-Affiliated Institute
- **Advancement Technology Development for Torpedo Deception Strategies in Submarines** (11/2022–11/2026) LIG Nex1
- **Advancement Tech Dev for Submarine Target Identification and Engagement Support Intelligence** (11/2022–11/2026) LIG Nex1
- **Research Planning for RL-based Scheduling in Autonomous Satellite Task Allocation** (05/2026–10/2026) KOITA
- **mmWave-based Liquid Classification in Next-Gen Robotic Vacuum Cleaners** (06/2026–09/2026) LG Electronics (Robotics Lab)
- **Mapping between Real World and VR for End-Edged Cloud Real-Time VR Servers** (09/2020–11/2025) Samsung Electronics
- **LEO Satellite Routing Research using Large Language Model and Reinforcement Learning** (05/2025–11/2025) ETRI
- **Discovering Quantum-Advantage Cases for Industry Adoption** (09/2025–11/2025) KTL
- **Research on Generative Quantum Machine Learning Models** (04/2025–10/2025) ETRI-Affiliated Institute
- **Quantum Hyper-Driving: Quantum-Inspired Hyper-Connected and Hyper-Sensing Autonomous Mobility** (03/2022–02/2025) NRF
- **Research on Learning-based Swarm Mission Planning Algorithms** (03/2024–02/2025) LIG Nex1
- **Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks** (05/2024–11/2024) ETRI
- **Korea-Japan Joint Seminar Project for Generative and Multi-Modal AI Technologies** (10/2023–09/2024) NRF
- **Quantum Machine Learning-based Objection Detection for Point Cloud and its Acceleration** (12/2022–04/2024) Hyundai Motors
- **NOMA-based Resource Allocation Research in Space-Air-Ground Integrated Networks** (09/2023–11/2023) ETRI
- **Routing Algorithms for LEO Satellite Networks** (12/2022–08/2023) Solvit System
- **Optimal Positioning Algorithms for Wide-Area Relaying Networks** (12/2022–08/2023) Solvit System
- **Distributed Learning Algorithms to Build AI Models with Multi-Center Clinical Data** (11/2022–02/2023) Cipherome
- **Cellular/Wi-Fi Handover Technology Development** (02/2022–12/2022) LG Electronics CTO Division (Smart Mobility Lab)
- **Autonomous Intelligent COA Search Methods for Cyber-Attacks** (12/2021–11/2022) ADD
- **Development of Quantum Deep Reinforcement Learning Algorithm using QAOA** (10/2019–04/2022) NRF
- **Research Trends in Digital Twin Applications to Autonomous Driving** (03/2022–04/2022) Hyundai NGV
- **Distributed Learning System Design and Implementation for Clinical Applications** (02/2022–03/2022) Cipherome
- **mmWave Radar and Deep Reinforcement Learning based Optimal Policy Autonomous Driving** (06/2019–02/2022) NRF
- **Research on Intelligent Agent-based CPS Security and Reliability** (04/2021–11/2021) TTA
- **Multi-GPU based Automotive HPC Platform Development** (04/2020–10/2020) ETRI
- **Super-Resolution Performance Optimization in Mobile Platforms** (05/2020–08/2020) Samsung SDS
- **Cooperative Deep Reinforcement Learning for Online Game Multi-Agents** (04/2020–08/2020) ETRI
- **Deep Learning Algorithms for mVOC Concentration Analysis** (03/2020–06/2020) Samsung Electronics (C-Lab)
- **Visual Recognition Software Implementation using Deep Learning Tools** (05/2019–11/2019) Hyundai Motors
- **Verification Testbed Implementation for Privacy-Preserving Trust Data Generation** (10/2019–11/2019) ETRI

- *Measurement and Analysis of Multi-Task GPU Scheduling Delays* (05/2019–10/2019) ETRI
- *mmWave High-Speed Networking Platform Design for Next-Generation Convergence Services* (06/2016–05/2019) NRF
- *Probabilistic Decision Making and Econometric Methods for Micro-Grid* (05/2017–04/2019) KEPCO Research Institute
- *GPU Scheduling Performance Analysis under Queueing Delay Considerations* (05/2018–10/2018) ETRI
- *Improving Massive Deep Learning Training via Computation and Communication Acceleration* (04/2018–10/2018) ETRI
- *A Priori Techniques Research for Efficient Multi-Edge Computing* (06/2017–12/2017) Samsung Electronics (Software Center)
- *Feasibility Study of 60 GHz IEEE 802.11ad for Virtual Reality (VR) Platforms* (04/2017–12/2017) IITP
- *Parsing Techniques for Artificial Neural Network (ANN) Data Processing* (09/2017–11/2017) ETRI

University of Southern California, Selected Projects

- *Video Aware Wireless Networks (VAWN) Research Program* Intel Labs, Verizon Wireless, Cisco Systems
- *60 GHz Real-Time Wireless Video Broadcasting* Disney Research Zürich
- *Annenberg Graduate Fellowship Award* University of Southern California

Korea University (KU), Selected Projects

- *Towards Agentic AI-Enabled Edge IoT Systems for Proactive Physical World Sensing and Control* (08/2025–07/2026) PKU-KU Joint Research Grant (Collaboration with Prof. Kaigui Bian at Peking Univ)
- *AI Teaching Assistant Research for LLM-based Large-Scale Education* (10/2024–02/2025) KU University College
- *Autonomous Mobility Control using Quantum Deep Learning* (03/2023–02/2024) KU Insung Research Grant
- *Mobile Access Algorithm Design using Economic Theory and AI* (03/2020–02/2021) KU Future Research Grant

Selected Publications

- **12,078+ Citations** (H-index: 54+, i10-index: 255+), obtained from Google Scholar Profile (as of July 5, 2026)
- **Journals and Magazines:** 173 publications; among them, 129 IEEE/Nature/APS publications

Books

- *Fundamentals of 6G Communications and Networking*, Springer (2024) (Editors: X. Lin, J. Zhang, Y. Liu, J. Kim)

Selected Papers

■ Top-Tier Conferences in AI (especially, Quantum AI)

- [ICML'26] *Quantum Robust Inner Minimization for Reinforcement Learning with Quadratic Speed-Up in Query Complexity*
ICML 2026 (H. K. Lee, J. Kim, S. W. Yoon) (Acceptance Rate: 26.56%)
- [ACL'26] *Let LLM Tutors Ask First: Proactive LLM-Based Tutoring at Scale in a 1,500-Student Online Classroom*
ACL 2026 (J. Lee, G. Youn, S. Lee, C. Im, J. Kim, C. Yoo)
- [CIKM'25] *Quantum-Amplitude Embedded Adaptation for Parameter-Efficient Fine-Tuning in Large Language Models*
ACM CIKM 2025 (E. J. Roh, J. Kim) (Acceptance Rate: 30.63%)
- [CIKM'25] *Filtered One-Shot Training for Quantum Architecture Search*
ACM CIKM 2025 (S. B. Son, S. Y.-C. Chen, J. Kim, S. Park) (Acceptance Rate: 30.63%)
- [CIKM'25] *LLM-based Interactive Coding Education via Predictive Query Management and Student-Centered Fine-Tuning: Design and Implementation with 1500-Student Class Data*
ACM CIKM 2025 (G. Youn, J. Lee, J. Kim, C. Yoo) (Acceptance Rate: 30.63%)
- [CIKM'24] *Hands-On Introduction to Quantum Machine Learning*
ACM CIKM 2024 (Tutorial) (S. Y.-C. Chen, J. Kim)
- [CIKM'23] *Quantum Split Learning for Privacy-Preserving Information Management*
ACM CIKM 2023 (S. Park, H. Baek, J. Kim) (Acceptance Rate: 27.44%)
- [CIKM'23] *Logarithmic Dimension Reduction for Quantum Neural Networks*
ACM CIKM 2023 (H. Baek, S. Park, J. Kim) (Acceptance Rate: 27.44%)
- [AAAI'23] *Quantum Multi-Agent Meta Reinforcement Learning*
AAAI 2023 (W. Yun, J. Park, J. Kim) (Acceptance Rate: 19.61%)
- [CIKM'22] *Hierarchical Reinforcement Learning using Gaussian Random Trajectory Generation in Autonomous Furniture Assembly*
ACM CIKM 2022 (W. Yun, D. Mohaisen, S. Jung, J.-K. Kim, J. Kim) (Acceptance Rate: 29.64%)
- [IJCAI'19] *Randomized Adversarial Imitation Learning for Autonomous Driving*
IJCAI 2019 (M. Shin, J. Kim) (Acceptance Rate: 17.89%)

■ Top-Tier Conferences in Networks, Mobility, and Systems (also related to Quantum Computing)

- [IPDPS'25] *AQUA: Hardware-Agnostic Qubit Allocation for Quantum Multi-Programming*
IEEE IPDPS 2025 (X. Piao, J. Shim, J. Kim, J. Kim) (Acceptance Rate: 24.71%)
- [WiOpt'25] *Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach*
IEEE/IFIP WiOpt 2025 (G. S. Kim, J. Chung, T. Q. Duong, S. Park, J. Kim)
- [WiOpt'24] *Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management*
IEEE/IFIP WiOpt 2024 (S. Lee, G. S. Kim, S. Park, J. Kim)
- [WiOpt'22] *Cooperative Video Quality Adaptation for Delay-Sensitive Dynamic Streaming using Adaptive Super-Resolution*
IEEE/IFIP WiOpt 2022 (M. Choi, W. Yun, J. Kim)
- [INFOCOM'22] *Joint Superposition Coding and Training for Federated Learning over Multi-Width Neural Networks*
IEEE INFOCOM 2022 (H. Baek, W. Yun, Y. Kwak, S. Jung, M. Ji, M. Bennis, J. Park, J. Kim) (Acceptance Rate: 19.93%)

- [ICDCS'20] *Understanding the Potential Risks of Sharing Elevation Information on Fitness Applications*
IEEE ICDCS 2020 (Ü. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen) (Acceptance Rate: 17.98%)
- [ICDCS'18] *ShmCaffe: A Distributed Deep Learning Platform with Shared Memory Buffer for HPC Architecture*
IEEE ICDCS 2018 (S. Ahn, J. Kim, E. Lim, W. Choi, A. Mohaisen, S. Kang) (Acceptance Rate: 20.63%)
- [ACM-MM'17] *REQUEST: Seamless Dynamic Adaptive Streaming over HTTP for Multi-Homed Smartphone under Resource Constraints*
ACM Multimedia 2017 (J. Koo, J. Yi, J. Kim, M. Hoque, S. Choi) (Acceptance Rate: 27.63%)
- [MobiSys'10] *Energy-Efficient Rate-Adaptive GPS-based Positioning for Smartphones*
ACM MobiSys 2010 (J. Paek, J. Kim, R. Govindan) (Acceptance Rate: 19.84%)

■ Magazines and Journals

◀ Accepted ▶

- [CIM.accepted] Rethinking Privacy-Preserving Distributed Learning in Quantum Hilbert Spaces, *IEEE Computational Intelligence Magazine*, v(n):ppp-ppp (S. B. Son, J. Kim, S. Park)
- [CM.accepted] Sustainable 6G Non-Terrestrial Networks: Enabling Net-Zero Operation via Renewable Energy Integration and Intelligent Control, *IEEE Communications Magazine*, v(n):ppp-ppp (H. Lee, S. Park, S. Jung, J. Kim)
- [CM.accepted] Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective, *IEEE Communications Magazine*, v(n):ppp-ppp (G. S. Kim, J. Chung, S. Park, S. Jung, T. Q. Duong, J. Kim)
- [NM.accepted] Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks, *IEEE Network*, v(n):ppp-ppp (H. Lee, G. S. Kim, E. J. Roh, S. Jung, S. Park, T. Q. Duong, J. Kim)
- [TMC.accepted] Scale-Reconfigurable Multi-Agent Reinforcement Learning for Aerial Networks in Dynamic Environments, *IEEE Transactions on Mobile Computing*, v(n):ppp-ppp (G. S. Kim, E. J. Roh, S. Jung, S. Park, J. Kim)
- [TMC.accepted] Efficient Quantum Neural Architecture Search with Filtered Supernet for Mobile Learning Systems, *IEEE Transactions on Mobile Computing*, v(n):ppp-ppp (S. B. Son, H. Ahn, S. Y.-C. Chen, S. Park, J. Kim)
- [TVT.accepted] Time-to-Collision Aware Autonomous Driving Motion Control under Risky Scenario Considerations, *IEEE Transactions on Vehicular Technology*, v(n):ppp-ppp (S. Kim, J. Kim, J. Kim, S. Jung)

◀ 2026 ▶

- [FGCS'26.09] GraMA: A Gradient Matrix-Guided Assignment Method for Solving Qubit Mapping Problems, *Future Generation Computer Systems*, 182:108485 (X. Piao, J. Kim, J.-K. Kim)
- [IOTJ'26.08] Learning-based Resilient Dynamic Routing for Fault-Tolerant Robust LEO Satellite Networks, *IEEE Internet of Things Journal*, 13(16):ppp-ppp (G. S. Kim, C. Im, Y. G. Kim, J. Ha, J. Kim)
- [EAAI'26.08] A Comprehensive Review on Quantum Deep Neural Networks for Prognostics and Health Management: Fundamentals, Challenges and Opportunities, *Engineering Applications of Artificial Intelligence*, 177:114991 (M. Jha, S. Chen, C. Kulkarni, J. Kim)
- [TMC'26.07] Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient Cube-Satellite Networks, *IEEE Transactions on Mobile Computing*, 25(7):10385–10401 (S. Park, G. S. Kim, Y. Cho, S. Jung, Z. Han, J. Kim)
- [Access'26.07] Two-Stage MPC-Guided Imitation and Reinforcement Learning for Autonomous Underwater Vehicle Control in Dynamic Obstacle Environments, *IEEE Access*, 14:ppp-ppp (S. Kim, E. J. Roh, I. Song, J. Kwon, S. Kwon, Y. Kim, J. Kim)
- [IOTJ'26.06] Quantum Federated Gradient Aggregation Using the Parameter-Shift Rule, *IEEE Internet of Things Journal*, 13(11):25367–25379 (J. Chung, C. Im, S. Park, J. Kim, W. Lee)
- [Nature'26.05] Fourier Analysis Perspective on Quantum Neural Networks, *Communications Physics (Nature)*, 9:176 (S. Oh, E. J. Roh, A. Vasilakos, S. Park, J. Kim)
- [TVT'26.04] Large-Scale Battery-Conscious Collision-Free Path-Finding for Sustainable and Autonomous Multi-AGV Mobility Control, *IEEE Transactions on Vehicular Technology*, 75(4):5364–5375 (Y. Cho, S. Ahn, S. Park, J. Kim)
- [TAES'26.03] Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket, *IEEE Transactions on Aerospace and Electronic Systems*, 62:3838–3863 (G. S. Kim, J. Chung, S. Jung, S. Park, J. Kim)
- [IOTJ'26.03] Gateway-Assisted Neural Angular Routing for Hierarchical Satellite Networks, *IEEE Internet of Things Journal*, 13(6):12322–12339 (J. Jang, I.-S. Cho, M. Shin, J. Kim, S. Jung)
- [IOTJ'26.02] Hybrid Large Language Model and Reinforcement Learning for Energy-Efficient Multisatellite Scheduling: Boosting the Performance from Scratch, *IEEE Internet of Things Journal*, 13(3):5379–5392 (H. Ahn, G. S. Kim, I.-S. Cho, S. Jung, J. Kim)
- [PRA'26.02] Understanding Qubit Trajectories in Parametrized Quantum Circuits, *Physical Review A*, 113:022609 (S. Oh, C. Im, S. Park, J. Kim)
- [AC'26.02] Collaborative Learning Architecture for Autonomous Excavator Planning and Execution, *Automation in Construction*, 182:106742 (J. Cho, M. Shin, J. Kim, S. Jung)
- [TAES'26.01] Integrated Control, Communication, and Computing for Mission-Critical Embedded Unmanned Aerial Vehicles, *IEEE Transactions on Aerospace and Electronic Systems*, 62:682–703 (G. S. Kim, S. Park, S. Jung, D. Mohaisen, J. Kim)
- [TMC'26.01] Quantum Multi-Agent Reinforcement Learning for Cooperative Mobile Access in Space-Air-Ground Integrated Networks, *IEEE Transactions on Mobile Computing*, 25(1):1200–1218 (G. S. Kim, Y. Cho, J. Chung, S. Park, S. Jung, Z. Han, J. Kim)
- ◀ 2025 ▶
- [MM'25.10-12] Quantum Jump to Virtual Worlds: High-Quality Multiple Virtual Meta-Space Realization in Metaverse, *IEEE MultiMedia*, 32(4):37–44 (S. Park, J. Kim)
- [TIV'25.11] Adaptive Quantum Federated Learning for Autonomous Surveillance Multi-Drone Networks, *IEEE Transactions on Intelligent Vehicles*, 10(11):5055–5059 (S. Park, C. Park, S. Jung, J. Kim)
- [JS'25.10] Joint Scalable Quantum Convolutional Neural Network and Reverse-Fidelity Training for High-Accurate Recognition in Unmanned Aerial Vehicle Surveillance, *The Journal of Supercomputing*, 81(16):1495 (E. J. Roh, J. Kim, S. Jung, S. Park)
- [TASE'25.09] Adaptive Excavation Automation in Complex Soil Environments using Reinforcement Learning, *IEEE Transactions on*

Automation Science and Engineering, 22:21181–21194 (M. Shin, J. Cho, J. Kim, S. Jung)

- [ESWA'25.09] Correlation-Assisted Spatio-Temporal Reinforcement Learning for Stock Revenue Maximization, *Expert Systems with Applications*, 289:128361 (J. Chung, M. Kim, S. Min, H. Choi, S. Park, J. Kim)
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Academic Activities and Research Supervision

Research Supervision

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- Dr. Hankyul Baek (03/2021–02/2024 (PhD), 03/2024–03/2025 (Postdoc)), ETRI AI Safety Institute (Researcher)
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- Dr. Gyu Seon Kim (03/2023–02/2026 (PhD), 03/2026–04/2026 (Postdoc)), RealtimeVisual (Researcher, *Military Service Exception*)
- Dr. Seok Bin Son (03/2022–02/2027 (Expected) (PhD))
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■ M.S. Alumni

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- Dohyun Kwon (03/2018–02/2020), Hyundai Marine & Fire Insurance (Researcher)
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IEEE Activities (Membership, Editorial Boards, and Services)

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 - *Associate Editor* (2025–), **ACM Computing Surveys**
 - *Associate Editor* (2025–), **IEEE Communications Surveys and Tutorials**
 - *Editor* (2023–), **IEEE Internet of Things Journal**
 - *Associate Editor* (2020–), **IEEE Transactions on Vehicular Technology**
 - *Guest Editor*, **Journal of Communications and Networks** (S.I. on Quantum Technologies for Communication Systems)
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